Disease Prediction System Report

Date: 2026-10-20 20:54

Patient Information

Name:	Marcus Vance	**Age:**	52
Gender:	Male	**Date:**	2026-10-20

Clinical Medical Report & Observations

Report Type: Comprehensive Metabolic Panel

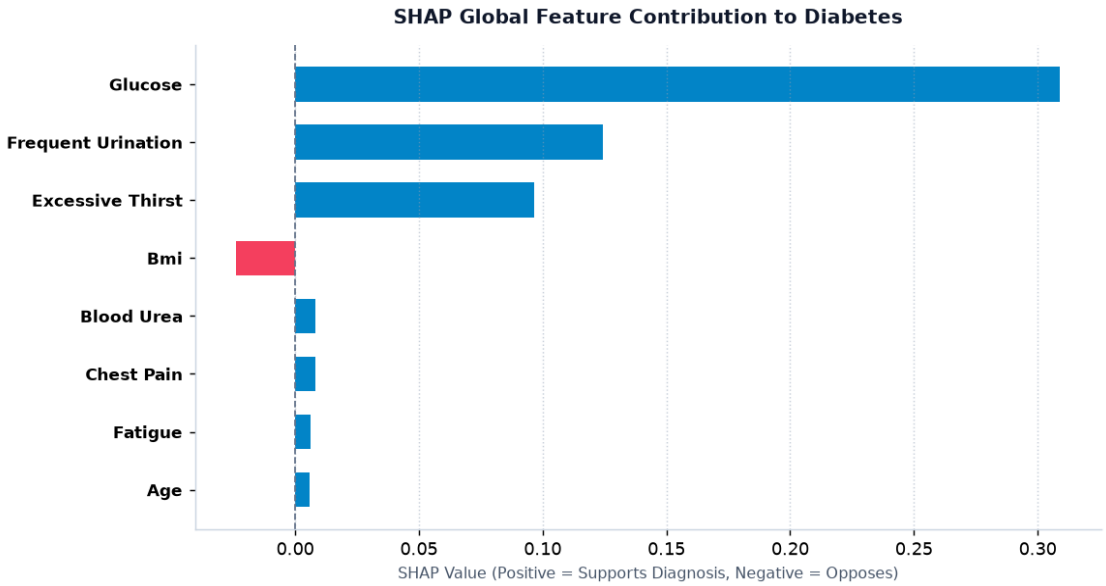
Fasting plasma glucose 175 mg/dL. HbA1c screening recommended.

Prediction Results

Disease	Probability
Diabetes	70.25%
Metabolic Syndrome	13.6%
Heart Disease	4.78%

Explainable AI: SHAP Global Feature Contribution

SHAP (SHapley Additive exPlanations) computes game-theoretic contribution values for each vital sign and clinical symptom.



Explainable AI: LIME Local Decision Rules

LIME (Local Interpretable Model-agnostic Explanations) builds a local surrogate linear model around this patient to identify the exact biometric thresholds driving the diagnosis.

